Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$a = 28$
			Total 1 mark

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms two equations of motion for 1.8 and 1.2 kg mass, three terms, condone sign error. "Whole string" approach scores M0	AO3.4	M1	$1.8 \times 9.81 - T = 1.8 \times a$
	Obtains two correct equations	AO1.1b	A1	$T - 1.2 \times 9.81 = 1.2 \times a$
	Solves to find correct value for a $0.2g$ or AWRT 1.96	AO1.1b	A1	$a = 1.96$ m s^{-2}
	Uses $s = ut + ½at^2$ with *their* calculated a value	AO1.1a	M1	$1.5 = ½ \times 1.96 \times t^2$
	Calculates correct t value AWRT 1.24	AO1.1b	A1	$t = 1.24$ seconds
(b)	Any valid assumption stated. Do not allow any comment already stated as an assumption in the question. Air resistance is permitted.	AO3.5b	E1	The string is long enough so that lighter mass does not reach the peg before the heavier mass hits the ground.
				Total 6 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States that the resistance	AO2.4	E1	Jason experiences less air

	is less for the second rider so the force required for equilibrium is also less, or mention of slipstreaming OE			resistance than Laura.
(b) (i)	Models situation by using given mass and total resistance force to form equation of motion PI	AO3.3	M1	As $F = ma$ then $a = -0.625$ m s^{-2}
	Uses appropriate suvat formula. May include $u = 25$.	AO3.4	M1	$u = 25$ km/h $= 6.944$ m s^{-1} $v = 0$; use $v^2 = u^2 + 2as$ so that $0 = 6.944^2 + 2 \times (-0.625) \times s$
	States correct value for s (AWFW 38.5 to 38.6) **Or** maximum $u = 7.07$ m s^{-1} or 25.5 km/h (AWRT) **Or** a needs to be < −0.603 (AWRT) **Or** Resistance needs to be > 38.5 N (AWFW 38.4 to 38.6) **Or** v^2 is −1.8 (AWRT) when $s = 40$	AO1.1b	A1	$s = 38.6$ m
	Makes appropriate comparison to conclude that Laura stops in time. Not necessary to see 38.6 < 40, but comparison for other variables must be clear.	AO3.2a	E1	So Laura stops before reaching the accident
(ii)	States an assumption that, if incorrect, would **contradict** the conclusion in (i). (eg reaction time, diminishing resistive force as speed drops OE)	AO3.5a	E1F	Taking account of reaction time would mean she travelled a distance before starting to brake.
				Total 6 marks

Q4.

	Marking Instructions	**AO**	**Marks**	**Typical Solution**
(a)	Shows integral of v with	AO1.1a	M1	$\int 0.06(2 + t - t^2)dt$

	Marking Instructions	AO	Marks	Typical Solution
	respect to t Condone omission of dt			
	Integrates with at least two correct terms (condone omission of constant at this stage) Any equivalent form.	AO1.1b	A1	$= 0.12t + 0.03t^2 - 0.02t^3 + c$
	Finds constant using $t = 0$ and displacement of 3. PI.	AO2.2a	M1	Using $t = 0$, $c = 3$
	Finds fully correct expression for r.	AO1.1b	A1	$r = 0.12t + 0.03t^2 - 0.02t^3 + 3$
(b)	Models problem as only force acting is that due to gravity so that $v = 0$ at highest point. (PI)	AO3.3	M1	$a = -9.8$ m s^{-2} $v = 0$ m s^{-1}
	Uses appropriate suvat formula $v = u + at$ with $u = 3.43$ and g negative or $u = -3.43$ and g positive	AO1.1a	M1	$0 = 3.43 - 9.8t_{max}$
	Finds $t_{max} = 0.35$ s (CAO) Condone addition of the 2 seconds to give $t = 2.35$ s or 2.4 s	AO1.1b	A1	$\therefore t_{max} = 0.35$ s
				Total 7 marks

Q5.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	0.0071 m s^{-2}
			Total 1 mark

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Forms equation of motion with four terms Condone sign error	AO3.4	M1	$300 - 140 - R = 482 \times 0.2$ $R = 63.6$ N
	Obtains correct equation.	AO1.1b	A1	
	Obtains correct value of R.	AO1.1b	A1	

(ii)	Forms equation of motion with correct terms Condone sign error	AO1.1a	M1	$T - 63.6 = 72 \times 0.2$ $T = 78$ N
	Obtains correct equation Follow through their *R*	AO1.1b	A1F	
	Obtains correct value of *T*	AO1.1b	A1	
(b)	States appropriate assumption **NOT friction or air resistance**	AO3.3	E1	Rope has no mass **or** is horizontal **or** is inextensible
(c) (i)	Forms equation of motion for skater using 'their' *R* Condone sign error	AO3.1b	M1	$-63.6 = 72a$
	Finds correct acceleration for 'their' *R*	AO1.1b	A1F	$a = -0.883\ldots$ m s^{-2}
	Uses a suitable constant acceleration formula with 'their' a	AO1.1a	M1	$u = 6$ $v = 0$ $a = -0.883$
	Obtains s when $v = 0$ Or Obtains v or positive v^2 when $s = 20$	AO1.1b	A1F	$0 = 6^2 - 2 \times 0.883s$ $s = 20.4$ m
	Explains that the skater hits buggy using correct values	AO3.2a	E1	20.4 > 20 Skater hits buggy
(ii)	Explains that the tension is removed from the buggy	AO2.4	E1	The rope is released so there is no tension acting on the buggy, so there is a higher resultant force. The driver will notice an increase in acceleration.
	Explains that the driver notices an increase in acceleration	AO2.4	E1	
				Total 14 marks

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds correct acceleration	AO1.1b	B1	0.5 m s^{-2}
(b)	Identifies $5T$ as the distance travelled after the first 15 seconds	AO3.4	B1	Distance at constant speed = $5T$ Distance in first 15 secs =

	Uses the information given to form an equation to find T (award mark for either trapezium expression separate, totalled or implied)	AO3.1b	M1	$\frac{1}{2} \times (3 + 8) \times 10 + \frac{1}{2} \times (8 + 5) \times 5$ $= 55 + 32.5 = 87.5$ $5T + 87.5 = 120$ So $T = 6.5$
	Correctly calculates the distance for the first 15 secs	AO1.1b	A1	
	Deduces the values of T from the mathematical models applied	AO2.2a	A1	
				Total 5 marks

Q8.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States correct expression for a	AO1.1b	B1	$a = \frac{V-U}{T}$
(b)	Rearranges to make T the subject of the formula	AO2.1	R1	$T = \frac{V-U}{a}$
	Uses given expression for S and attempts to eliminate T	AO2.1	R1	$S = \frac{1}{2}(U+V) \times \frac{V-U}{a}$ $2aS = (U + V)(V - U)$
	Completes argument to reach required result **AG** Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$V^2 = U^2 + 2aS$ **(AG)**
				Total 4 marks

Q9.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates two (or four) appropriate distances	AO3.1b	M1	$s_1 = \frac{1}{2}(6 + 10) \times 8 = 64$ m

	Obtains correct distances	AO1.1b	A1	$s_2 = \frac{1}{2} \times 10 \times 2 = 10$ m
	Obtains correct sum of 'their' distances	AO1.1b	A1F	$s_1 + s_2 = 64 + 10 = 74$ m **OR** $S_1 = 6 \times 8 = 48$ m $S_2 = \frac{1}{2} \times 4 \times 8 = 16$ m $S_3 = \frac{1}{2} \times 4 \times 2 = 4$ m $S_4 = \frac{1}{2} \times 6 \times 2 = 6$ m $S_1 + S_2 + S_3 + S_4 = 74$ m
(b)	Finds difference of 'their' distances from part **(a)**	AO2.2a	B1F	$64 - 10 = 54$ m
(c)	Calculates magnitude of acceleration	AO3.1b	M1	$a_{\text{max}} = \frac{8}{4} = 2$
	Obtains correct resultant force	AO1.1b	A1	$F_{\text{max}} = 800 \times 2 = 1600$ N
(d)	Explains that abrupt changes and straight lines in the graph are unlikely in reality	AO3.5b	E1	Change of velocity is unlikely to result in abrupt changes I would expect to see curves on the graph
				Total 7 marks

Q10.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct horizontal component of the initial velocity	AO1.1b	B1	$2.5U = 40$ $U = 16$
	Forms equation to find vertical component of initial velocity	AO3.3	M1	$-10 = 2.5V - 0.5 \times 9.81 \times 2.5^2$
	Obtains correct vertical component of initial velocity	AO1.1b	A1	$V = 8.2625$
	Forms equation for vertical component of velocity at height 3 using 'their'	AO3.4	M1	$v_y^2 = 8.2625^2 + 2 \times (-9.81) \times 3$

	derived values for U and V			
	Obtains correct component of velocity	AO1.1b	A1	$v_y = 3.067\ldots$
	Correct final speed with units, correct for 'their' U and v_y FT applies only if both M1 marks have been awarded	AO3.2a	A1F	$v = \sqrt{16^2 + 3.067^2} = 16.3\text{ m s}^{-1}$
(b)	States 'their' value of horizontal component of the initial velocity from part **(a)**	AO3.4	A1F	16 m s $^{-1}$
(c)	Explains that horizontal velocity has been assumed to be constant in their model and that this is not likely to be true, with valid reasoning	AO3.5b	E1	It was assumed that there were no resistance forces acting on the ball which is unlikely to be true in reality. The horizontal speed of the ball is likely to vary… air resistance would slow the ball down, wind might speed the ball up
				Total 8 marks

Q11.

(a) (i) $640 = \frac{1}{2}(12 + 20)t$

M1: Use of constant acceleration equation to find t with s = 640, 20 and 12.

A1: Correct equation.

M1A1

$t = \frac{640 \times 2}{32} = 40$ s

A1: Correct time.

A1

For two equation methods, award no marks until an equation for t is obtained.

Using a = 0.2 to find t = –40 scores M1A0A0

3

(ii) $12^2 = 20^2 + 2 \times a \times 640$

M1: Use of constant acceleration equation to find a with u = 20 and v = 12.

A1F: Correct equation.

M1A1

$a = \dfrac{12^2 - 20^2}{2 \times 640} = -0.2\text{ m s}^{-2}$

(Deceleration = 0.2 m s^{-2})

A1F: Correct deceleration.

Do not award for a = 0.2

Accept –0.2 or ± $\frac{1}{5}$ m s^{-2} for deceleration

A1

3

OR

$12 = 20 + 40a$

(M1A1F)

$a = \dfrac{-8}{40} = -0.2\text{ m s}^{-2}$

(Deceleration = 0.2 m s^{-2})

(A1F)

(3)

OR

$640 = 20 \times 40 + \frac{1}{2}\, a \times 40^2$

(M1A1F)

$a = \dfrac{-160}{800} = -0.2\text{ms}^{-1}$

(Deceleration = 0.2 m s^{-2})

(A1F)

(3)

Follow through incorrect times from part (a).

Accept $\frac{8}{40}$ = 0.2 provided that the equations
20 = 12 + 40 a or $20^2 = 12^2 + 1280a$ are not seen

$a = \frac{8}{40}$ = 0.2 scores M1A1A0 unless a is defined as deceleration

(b) (i) $1820 = 12 \times 70 + \frac{1}{2} \times a \times 70^2$

M1: Constant acceleration equation to find a with
u = 12 (or 20),
s = 1820 and t = 70.
A1F: Correct equation.

M1A1

$a = \frac{1820 - 21 \times 70}{2450} = 0.4 \text{ m s}^2$

A1F: Correct acceleration. Accept $\frac{2}{5}$ m s^{-2} oe.

A1

3

(ii) $1820 = \frac{1}{2}(12 + v) \times 70$;

M1: Constant acceleration equation to find v with
s = 1820 and t = 70.
A1F: Correct equation.

M1A1

$v = \frac{1820}{35} - 12 = 40 \text{ m s}^{-1}$

A1F: Correct velocity.

A1

3

OR

$v = 12 + 0.4 \times 70$

(M1A1F)

$= 40 \text{ m s}^{1\square}$

(A1F)

(3)

OR

$v^2 = 12^2 + 2 \times 0.4 \times 1820$

(M1A1F)

$v = \sqrt{1600} = 40 \text{ m}^{1\square}$

(A1F)
(3)

OR

$1820 = 70\,v - \frac{1}{2} \times 0.4 \times 70^2$

(M1A1F)

$v = 40$ m s^{-1}

(A1F)
(3)

For two equation methods, award no marks until an equation for v is obtained.

(c) Average Speed = $\frac{640+1820}{40+70}$

M1: Division of 2460 by their total time (70 + their answer to (a)).

M1

$= \frac{2460}{110} = 22.4$ m s^{-1}

A1F: Correct time. Accept 22.3 or AWRT 22.4

A1F
2

[14]

Q12.

(a) $\mathbf{v} = (6\mathbf{i} + 2.4\mathbf{j}) + (-0.8\mathbf{i} + 0.1\mathbf{j})t$

M1: Using constant acceleration equation to get v.

A1: Correct expression for the velocity. Allow equivalent column vector answer.

M1A1
2

(b) $\mathbf{r} = (6\mathbf{i} + 2.4\mathbf{j})t + \frac{1}{2}(-0.8\mathbf{i} + 0.1\mathbf{j})t^2 + 13.6\mathbf{i}$

M1: Use of $\mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$ at or other constant acceleration equation.

*A1: Position vector with or without 13.6**i**.*

M1A1

$(= (6t - 0.4t^2 + 13.6)\mathbf{i} + (2.4t + 0.05t^2)\mathbf{j})$

A1: Correct position vector.

A1

3

(c) $\mathbf{v} = (6 - 0.8t)\mathbf{i} + (2.4 + 0.1t)\mathbf{j}$

B1: Velocity simplified into $\boldsymbol{i}$ *and* $\boldsymbol{j}$ *components. Could be implied.*

B1

$6 - 0.8t = -(2.4 + 0.1t)$

M1: $6 - 0.8t = \pm(2.4 + 0.1t)$

A1: Correct equation.

M1A1

$8.4 = 0.7t$

$t = \frac{8.4}{0.7} = 12$ s

A1: Correct t.

A1

$\mathbf{r} = 28\mathbf{i} + 36\mathbf{j}$

dM1: Finding position vector using their time.

A1: Correct position vector.

dM1A1

$d = \sqrt{28^2 + 36^2} = 45.6$m

A1: Correct distance. Accept AWRT 45.6

A1

Do not penalise the use of other methods, such as trial and improvement, to find the time.

7

[12]

Q13.

(a) $s = \frac{1}{2}(5 + 4) \times 6 + \frac{1}{2}(4 + 7) \times 8 + 7 \times 7$

M1: Method based on three (or four or more!) areas / distances or equivalent added together.

A1: Correct calculation or value for one area / distance for one time period (eg 0 to 6 seconds).

M1A1

= 27 + 44 + 49

A1: Correct calculation or value for area / distance for another time period.

A1

= 120 m

A1: Correct final distance.

A1

For example 24 + 44 + 49 = 117 scores M1A1A1A0.

4

(b) Average Speed = $\frac{120}{21}$ = 5.71 m s^{-1}

M1: Their answer to part (a) divided by 21.

M1

A1F: Correct average speed.

Accept $5\frac{5}{7}$ or $\frac{40}{7}$.

A1F

2

[6]

Q14.

(a) $3g - T = 3a$

M1: Three term equation of motion with $3g$ or 29.4, T and $3a$.
A1: Correct equation.

M1A1

$T - g = a$

$2g = 4a$

M1: Three term equation of motion with g or 9.8, T and a.
A1: Correct equation.

M1A1

$a = \frac{g}{2} = 4.9$ m s^{-2}

A1: Correct final answer. Accept $\frac{g}{2}$

A1

Note: Do not penalise candidates who consistently use *signs in the opposite direction throughout, provided they then give their final answer as 4.9, having seen –4.9 in their working. If the final answer is –4.9 don't award the final A1 mark.*

Special Case:

Whole string method $2g = 4a$ *and* $a = \frac{2g}{4} = 4.9$ *OE scores M1A1A1*

5

(b) $v^2 = 0^2 + 2 \times 4.9 \times 0.4$

M1: Use of a constant acceleration equation to find v*, with* $u = 0$*, their value for a from part (a) and* $s = 0.4$ *or 40.*

M1

$v = \sqrt{3.92} = 1.98 \text{ m s}^{-1}$

A1F: Correct speed. Follow through their acceleration from part (a).

Use $v = \sqrt{0.8a}$ *for FT.*

Accept 1.97.

If candidates use two equations, award no marks until they have an equation for v*. (Note use of* $t = 0.404$ *or better required for A1)*

A1F

2

(c) $0^2 = \left(\sqrt{3.92}\right)^2 + 2 \times (-9.8)\, s$

M1: Use of a constant acceleration equation with $v = 0$*,* $a = \pm 9.8$ *and their speed from (b).*

M1

$s = \frac{3.92}{2 \times 9.8} = 0.2 \text{ m}$

A1: Correct distance.

A1

Total = 0.2 + 0.4 = 0.6 m

A1: Correct total distance. Allow 60 cm from correct working.

Note

$0^2 = \left(\sqrt{392}\right)^2 + 2 \times (-9.8)s$

$s = \frac{392}{2 \times 9.8} = 20$

scores M1A0A0

A1

If candidates use two equations, award no marks until they have an equation for s. (Note use of t = 0.202 or better required for A marks)

3

(d) The acceleration would be less, because the resultant force on each particle would be reduced.

B1: Less
'Slower acceleration' not acceptable

B1

B1: Appropriate reason.
Only award second B1 if they say acceleration is less.

B1

2

[12]

Q15.

(a) $\mathbf{r} = (-17.5\mathbf{i} - 27\mathbf{j})t + \frac{1}{2}(0.5\mathbf{i} + 0.6\mathbf{j})t^2 + (500\mathbf{i} + 200\mathbf{j})$

M1: Use of $\mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$
A1: Correct with or without the initial position. That is with the final term missing or on the wrong side.

M1A1

A1: Correct with the initial position included.

A1

OR

$\mathbf{r} = (500 - 17.5t + 0.25t^2)\mathbf{i} + (200 - 27t + 0.3t^2)\mathbf{j}$

3

(b) $200 = -17.5t + 0.25t^2 + 500$

M1: Forming equation for one component based on position of the rock and their position vector.

$0.25t^2 - 17.5t + 300 = 0$

A1: Correct quadratic equation.

M1A1

$t = 40$ or 30

A1: At least one correct solution.

A1

$-400 = -27t + 0.3t^2 + 200$

dM1: Forming equation for the other component.

dM1

$0.3t^2 - 27t + 600 = 0$

A1: Correct equation.

A1

$t = 40$ or 50

dM1: Obtaining one or two positive solutions.

dM1

$\therefore t = 40$

A1: Selecting 40.

A1

7

OR

$-27 \times 40 + 0.3 \times 40^2 + 200 = -1080 + 480 + 200$

dM1: Substituting 40 into the other component.

(dM1)

A1: Correct substitution

(A1)

$= -400$

dM1: Checking this component of the position vector

(dM1)

$\therefore t = 40$

A1: Concluding that $t = 40$

Note that alternative methods based on trial and improvement can be awarded full marks.

(A1)

Alternative methods

$0.25t^2 - 17.5t + 300 = 0$

Marks allocated as above

(M1A1)

$0.3t^2 - 27t + 600 = 0$

(dM1A1)

$0.55t^2 - 44.5t + 900 = 0$

$t = 40$ or $t = 40.9$

A1: At least one correct solution

(A1)

$0.25 \times 40^2 - 17.5 \times 40 + 300 = 0$

dM1: Checking one or both solutions

(dM1)

$0.3 \times 40^2 - 27 \times 40 + 600 = 0$

$\therefore t = 40$

A1: concluding $t = 40$

(A1)

$0.05t^2 - 9.5t + 300 = 0$

$t = 40$ or $t = 150$

A1: At least one correct solution

(A1)

$0.25 \times 40^2 - 17.5 \times 40 + 300 = 0$

dM1: Checking one or both solutions

(dM1)

$0.3 \times 40^2 - 27 \times 40 + 600 = 0$

$\therefore t = 40$

(A1)

(7)

(c) Av. Velocity = $\dfrac{(200\mathbf{i} - 400\mathbf{j}) - (500\mathbf{i} + 200\mathbf{j})}{40}$

$= \dfrac{-300\mathbf{i} - 600\mathbf{j}}{40}$

M1: Use of change in position over time, with a subtraction to obtain position. Do not award if one position is taken as the origin.

M1

A1F: Correct expression.

A1F

$= -7.5\mathbf{i} - 15\mathbf{j}$

A1F: Correct final answer.

A1F

Follow through on their time from part (b).

Av Vel = $\dfrac{-300\mathbf{i} - 600\mathbf{j}}{t}$

3

(d) No – The helicopter will follow a curved path and not move along a straight line between the two positions.

B1: No.
B1: Mentions path is longer than the distance between the two points.
Only award second B1 if the candidate has stated that the two quantities are not equal.

B2,1

2

[15]

Q16.

(a) $x = ut\cos\theta$

M1

$t = \dfrac{x}{u\cos\theta}$

A1

$y = -\dfrac{1}{2}gt^2 + ut\sin\theta$

Condone + g for M1

M1

$y = -\dfrac{1}{2}gt^2 + ut\sin\theta$

A1

$y = -\dfrac{1}{2}g\left(\dfrac{x}{u\cos\theta}\right)^2 + u\left(\dfrac{x}{u\cos\theta}\right)\sin\theta$

Elimination of t, condone + g for m1

m1

$$y = -\frac{gx^2}{2u^2\cos^2\theta} + \frac{x\sin\theta}{\cos\theta}$$

$$y = -\frac{gx^2}{2u^2}(1 + \tan^2\theta) + x\tan\theta$$

OE in terms of x, u, g, $\tan\theta$

A1

6

(b) (i) $0.5 = -\frac{9.8(5)^2}{2(8)^2}(1 + \tan^2\theta) + 5\tan\theta$

Correctly substituting for x, y, u and g into their equation of trajectory

M1

All correct, condone decimal approximation.

A1

$245\tan^2\theta - 640\tan\theta + 309 = 0$

OE exact quadratic in $\tan\theta$

A1

$$\tan\theta = \frac{640 \pm \sqrt{(-640)^2 - 4(245)(309)}}{2(245)}$$

PI by the values of $\tan\theta$

m1

$\tan\theta = 1.973(004),\quad 0.6392(41)$

$\theta = 63.12°,\ 32.58°$

$\theta = 63.1°,\ 32.6°$

AG Must see the above or more accurate values

A1

5

(ii) $\dot{y} = -9.8\left(\frac{5}{8\cos 63.1°}\right) + 8\sin 63.1°$ OE

Condone +9.8 for M1.

M1

($\dot{y} = -6.4035$)

$\dot{x} = 8\cos 63.1°$

($\dot{x} = 3.6195$)

M1

$\tan^{-1} \frac{6.4(035)}{3.6(195)}$ (= 61°) OE

PI by correct angle in a statement

m1

Direction: 61° to the horizontal
or 29° to the vertical

Have to see "horizontal" or "vertical" or diagram

A1

4

Alternative:

$\frac{dy}{dx} = -\frac{2gx}{2u^2}(1 + \tan^2\theta) + \tan\theta$

(M1)

$= -\frac{2 \times 9.8 \times 5}{2 \times 8^2}(1 + \tan^2 63.1°) + \tan 63.1°$

$= -1.7692$

(A1)

$\tan^{-1}(-1.7692) = -60.52368°$

(m1)

Direction: 61° to the horizontal
or 29° to the vertical

(A1)

(4)

(c) The ball is a particle, or
No air resistance, or

The ball does not spin

B1

1

[16]

Q17.

(a) $s_1 = \frac{1}{2} \times 5 \times 28 = 70$ m

M1: For $\frac{1}{2} \times 5 \times 28$ or equivalent.
A1: Correct distance

M1A1

2

(b) $s = 70 + \frac{1}{2} \times 5 \times 22$

B1: For $\pm \frac{1}{2} \times 5 \times 22$ or equivalent.
M1: For adding the distances.

B1M1

= 70 + 55

= 125 m

A1F: Correct distance. Follow through their answer from part (a) only.

A1F

3

(c) Average speed = $\frac{125}{50} = 2.5$ ms^{-1}

M1: For their answer to (b) divided by 50.

M1

A1F: Correct average speed. Follow through answers from part (b).

A1F

2

(d) Displacement from $O = 70 - 55 = 15$ m

B1: Correct displacement.

B1

1

(e) Average velocity = $\frac{15}{50}$ = 0.3 ms^{-1}

M1: For their answer to (d) divided by 50, provided they have subtracted in (d).

M1

A1F: Correct average velocity. Follow through answers from part (d).

Award no marks if the final answer is 0.

A1F

2

(f) $a = \frac{5}{18}$ = 0.278 ms^{-2}

B1: Correct acceleration. Accept $\frac{5}{18}$ or equivalent fraction or 0.277 or AWRT 0.278.

Condone 0.28.

B1

1

[11]

Q18.

(a) (i) $10^2 = 4^2 + 2 \times a \times 50$

M1: Use of a constant acceleration equation to find a, with v and u substituted correctly.

For example $4^2=10^2 + 100a$ scores M0A0A0.

A1: Correct constant acceleration equation.

M1A1

$a = \frac{100-16}{100}$ = 0.84 ms^{-2}

A1: Correct a.

A1

Note if t found first award M1 for use of

$v = u + at$ *or* $s = ut + \frac{1}{2}at^2$.

3

(ii) $50 = \frac{1}{2}(4 + 10)t$

M1: Use of a constant acceleration equation to find t.

A1F: Correct constant acceleration equation with their acceleration from (a)(i) seen.

M1A1

$t = \frac{50}{7} = 7.14$ s

A1: Correct t. Accept $\frac{50}{7}$ or $7\frac{1}{7}$ or AWRT 7.14.

A1

OR

$10 = 4 + 0.84t$

If t has been found in part (a) the working does not have to be repeated, but value of t must be stated.

(M1A1F)

$t = \frac{6}{0.84} = 7.14$ s

Do not follow through incorrect values of a.

(A1)

OR

$50 = 4t + \frac{1}{2} \times 0.84t^2$

(M1A1F)

$0.42t^2 + 4t - 50 = 0$

$t = 7.14$ (or $t = -16.6$)

(A1)

3

(b) $70 \times 0.84 = 58.8$ N

M1: Use of $F = ma$ with $m = 70$ and their a from (a)(i).

A1F: Correct F. Follow through their value of a from part (a)(i).

M1A1F

2

(c) (i) $58.8 = 70 \times 9.8 \sin \alpha$

$\sin \alpha = \frac{58.8}{70 \times 9.8} = 0.08571$

M1: Resolving parallel to the slope must see $70g$ or

mg OE with sin α or cos α and their answer to part (b).
A1F: Correct equation. Follow through their answer to part (b) provided sin α < 1

M1A1F

$\alpha = 4.92°$

A1F: Correct angle. Follow through their answer to part (b). Accept 4.91° provided sin α < 1.

A1F

3

(ii) $70 \times 9.8 \sin \alpha - 30 = 58.8$

$\sin \alpha = 0.12945$

M1: Three term equation of motion. Must see 70g or mg OE with sin α or cos α.
A1F: Correct equation. Follow through their answer to part (b) provided sin α < 1

M1A1F

$\alpha = 7.44°$

A1F: Correct angle. Follow through their answer to part (b) provided sin α < 1.
Accept 7.43°.
Accept 7.41° from 0.129.

A1F

3

(d) The air resistance force will increase (vary or change) with speed.

B1: Correct statement.

B1

1

[15]

Q19.

(a) $10^2 = 20^2 + 2 \times a \times 75$

M1: Use of a constant acceleration equation to find a, with v = 10 and u = 20.
$20^2 = 10^2 + 2 \times a \times 75$ scores M0
A1: Correct equation.

M1A1

$a = \dfrac{100-400}{150} = -2\ \text{ms}^{-2}$

A1: Correct acceleration.
For two equation methods award no marks until an equation for a is obtained.

A1

3

(b) $0 = 20 - 2t$

M1: Using a constant acceleration equation, with u = 20 and v = 0, to find t using their acceleration from (a) even if positive.
Using s = 75 scores M0

M1

t = 10 seconds

A1: Correct time from correct working CSO.

A1

2

(c) $F = 1400 \times 2 = 2800$ N

M1: Use of $F = ma$ with ± their acceleration and mass of 1400.

A1F: Correct force. Follow through the magnitude of their acceleration. Answer must be positive. Sign changes do not need to be justified.

M1A1F

2

[7]

Q20.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

M1: Using constant acceleration equation to get $\mathbf{r}$.

A1: Correct expression for $\mathbf{r}$. Allow equivalent column vector answer.

M1A1

2

(b) $3t - 0.1t^2 = 0$

$t(3 - 0.1t) = 0$

$t = 0$ or $t = 30$

M1: Putting their $\boldsymbol{j}$ component equal to zero to form a quadratic equation.
A1: Correct equation.

M1A1

t = 30 seconds

A1: For 30 seconds. No need to see t = 0.

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

B1: Correct expression for the velocity in terms of t.
Can be implied by subsequent working in terms of t.

B1

$0.1t - 1 = -(3 - 0.2t)$

$2 = 0.1t$

M1: For $0.1t - 1 = \pm(3 - 0.2t)$. May be with their components if velocity stated incorrectly.
A1: Correct equation.

M1A1

$t = 20$

A1:t = 20

A1

$\mathbf{v} = \mathbf{i} - \mathbf{j}$

$v = \sqrt{2} = 1.41 \text{ ms}^{-1}$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases
If the equation in t in line 2 is not seen:
then seeing t = 20 and $\boldsymbol{v} = \boldsymbol{i} - \boldsymbol{j}$ and v = 1.41 award 4 out of 6
or
then seeing t = 20 and $\boldsymbol{v} = \boldsymbol{i} - \boldsymbol{j}$ award 2 out of 6

6

[11]

Q21.

(a) $22.4 \sin\theta - 2 \times 9.8 = 0$

M1: Use of $v = u + at$ vertically with $u = 22.4\sin\theta$, $v = 0$, $t = 2$ and $a = \pm 9.8$.

A1: Correct equation. (May be in terms of g or contain 9.81..

M1A1

$\sin\theta = \frac{19.6}{22.4} = \frac{7}{8} = 0.875$

A1: Must see either

$22.4 \sin\theta = 19.6$ or $\frac{19.6}{22.4}$

A1

OR

M1: Use of $s = ut + \frac{1}{2}at^2$ with $u = 22.4\sin\theta$, $s = 0$, $t = 4$ and $a = \pm 9.8$.

A1: Correct equation.

A1: must see $89.6 \sin\theta = 78.4$ or $\frac{78.4}{89.6}$ OE

3

(b) $h_{MAX} = 22.4 \times \frac{7}{8} \times 2 - \frac{1}{2} \times 9.8 \times 2^2$

M1: Using a constant acceleration equation to find height, with $t = 2$, $u = 22.4 \sin\theta$ or 19.6 and $a = \pm 9.8$.

A1: Correct equation.

M1A1

= 19.6 m

A1: Correct height. AWRT 19.6

Note using $g = 9.81$ gives 19.6, also accept 19.5.

Note: other constant acceleration equations will lead to the same result

A1

3

(c) $\cos\theta = \frac{\sqrt{15}}{8} = 0.4841$ or $\theta = 61.04°$

B1: Correct value for $\cos\theta$ (accept 0.484) or θ (accept 61.0° or 61° or 1.06 or 1.065 or 1.07 radians). Can be implied.

B1

$AB = 22.4 \times \frac{\sqrt{15}}{8} \times 4 = 43.4$ m

M1: Calculation for range with value for cosθ and with t = 4.

A1F: Correct distance. Follow through incorrect θ. Accept AWRT 43.4 or 43.3 or 43.2.

Do not accept 43.

M1A1F

3

(d) $22.4 \times \frac{7}{8}t - 4.9t^2 = 5$

M1: Use of $s = ut + \frac{1}{2}at^2$ with correct terms, but not necessarily signs.

M1

$4.9t^2 - 19.6t + 5 = 0$

A1: Correct equation.

A1

$t = 0.274$ or $t = 3.726$

dM1: Solving their quadratic.

dM1

A1: At least one correct solution. Allow 0.27 or 0.28 and 3.72 or 3.73

A1

Time = 3.726 – 0.274 = 3.45 seconds

A1: Correct difference. Accept 3.46.

A1

Note: there are other methods which will lead to the correct time:
M1dM1A1 for a constant acceleration equation that gives a time or times from which the final answer can be obtained

A1 Correct time or times

A1 Correct final answer

5

[14]

Q22.

(a) (i) $10^2 = 20^2 + 2 \times a \times 75$

M1: Use of a constant acceleration equation to find a, with v = 10 and u = 20.
$20^2 = 10^2 + 2 \times a \times 75$ scores M0
A1: Correct equation.

M1A1

$a = \dfrac{100-400}{150} = -2 \text{ ms}^{-2}$

A1: Correct acceleration.

A1

For two equation methods award no marks until an equation for a is obtained.

3

(ii) $0 = 20 - 2t$

M1: Using a constant acceleration equation, with u = 20 and v = 0, to find t using their acceleration from (a) even if positive.
Using s = 75 scores M0

M1

t = 10 seconds

A1: Correct time from correct working CSO.

A1

2

(iii) $F = 1400 \times 2$

M1: Use of $F = ma$ with ± their acceleration and mass of 1400.

M1

= 2800 N

A1F: Correct force. Follow through the magnitude of their acceleration. Answer must be positive. Sign changes do not need to be justified.

A1F

2

(b) F = 2800 – 200 = 2600 N

B1F: The magnitude of their force minus 200.
Do not award if M1 not awarded in (a)(iii).
Final answer must be positive.

Follow through only if their answer to (a)(iii) is greater than 200.

B1F

1

[8]

Q23.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

M1: Using constant acceleration equation to get $\boldsymbol{r}$*.*

A1: Correct expression for $\boldsymbol{r}$*. Allow equivalent column vector answer.*

M1A1

2

(b) $3t - 0.1t^2 = 0$

$t(3 - 0.1t) = 0$

$t = 0$ or $t = 30$

M1: Putting their $\boldsymbol{j}$ *component equal to zero to form a quadratic equation.*

A1: Correct equation.

M1A1

$t = 30$ seconds

A1: For 30 seconds. No need to see $t = 0$*.*

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

B1: Correct expression for the velocity in terms of t.
Can be implied by subsequent working in terms of t.

B1

$0.1t - 1 = -(3 - 0.2t)$

$2 = 0.1t$

M1: For 0.1t – 1 = ±(3 – 0.2t). May be with their components if velocity stated incorrectly.

A1: Correct equation.

M1A1

$t = 20$

A1: t = 20

A1

$\mathbf{v} = \mathbf{i} - \mathbf{j}$

$v = \sqrt{2} = 1.41 \text{ ms}^{-1}$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases
If the equation in t in line 2 is not seen:
then seeing t = 20 and $\boldsymbol{v} = \boldsymbol{i} - \boldsymbol{j}$ and v = 1.41
award 4 out of 6
or
then seeing t = 20 and $\boldsymbol{v} = \boldsymbol{i} - \boldsymbol{j}$ award 2 out of 6

6

[11]

Q24.

(a) 22.4 sin θ – 2 × 9.8= 0

M1: Use of v = u + at vertically with
u = 22.4 sin θ , v = 0, t = 2 and a = ±9.8.
A1: Correct equation. (May be in terms of g or contain 9.81..

M1A1

$\sin\theta = \frac{19.6}{22.4} = \frac{7}{8} = 0.875$

A1: Must see either

22.4 sin θ = 19.6 or $\frac{19.6}{22.4}$
AG

A1

Or

$0 = 22.4 \sin\theta \times 4 - \frac{1}{2} \times 9.8 \times 4^2$

M1: Use of $s = ut + \frac{1}{2}at^2$ with u = 22.4sinθ , s = 0, t = 4 and a = ±9.8.
A1: Correct equation.

(M1A1)

$\sin\theta = \frac{4.9\times16}{22.4\times4} = 0.875$

A1: must see 89.6 sin θ = 78.4 or $\frac{78.4}{89.6}$ *OE*

(A1)

3

(b) $h_{MAX} = 22.4 \times \sin\theta \times 2 - \frac{1}{2} \times 9.8 \times 2^2$

M1: Using a constant acceleration equation to find height, with t = 2, u = 22.4 sin θ or 19.6 and a = ±9.8.

A1: Correct equation.

M1A1

= 19.6 m

A1: Correct height. AWRT 19.6

Note using g = 9.81 gives 19.6, also accept 19.5.

Note: other constant acceleration equations will lead to the same result

A1

Or

$0^2 = (22.4 \times \sin\theta)^2 + 2 \times(-9.8)h_{MAX}$

(M1A1)

$h_{MAX} = 19.6\text{m}$

(A1)

3

(c) $\cos\theta = \frac{\sqrt{15}}{8} = 0.4841$ or $\theta = 61.04°$

B1: Correct value for cosθ (accept 0.484) or θ (accept 61.0° or 61° or 1.06 or 1.065 or 1.07 radians). Can be implied.

B1

$AB = 22.4 \times \frac{\sqrt{15}}{8} \times 4 = 43.4$ m

M1: Calculation for range with value for cosθ and with t = 4.

A1F: Correct distance. Follow through

incorrect θ. Accept AWRT 43.4 or 43.3 or 43.2.

Do not accept 43.

M1A1F

3

(d) $22.4 \times (\sin \theta)t - 4.9t^2 = 5$

M1: Use of $s = ut + \frac{1}{2}at^2$ with correct terms, but not necessarily signs.

M1

$4.9t^2 - 19.6t + 5 = 0$

A1: Correct equation.

A1

$t = 0.274$ or $t = 3.726$

dM1: Solving their quadratic.

dM1

A1: At least one correct solution. Allow 0.27 or 0.28 and 3.72 or 3.73

A1

Time = 3.726 – 0.274 = 3.45 seconds

A1: Correct difference. Accept 3.46.

A1

Note: there are other methods which will lead to the correct time:

M1dM1A1 for a constant acceleration equation that gives a time or times from which the final answer can be obtained

A1 Correct time or times

A1 Correct final answer

5

(e) $v_{MIN} = 22.4 \times \cos \theta$

M1: Finding horizontal component with candidate's value for $\cos \theta$. Do not award if combined with a non-zero vertical component.

M1

$= 10.8 \text{ m s}^{-1}$

A1: Correct speed. Accept 10.9 or 10.85.

A1

Or

$v_{min} = \frac{43.4}{4}$ = 10.9 to 3sf

M1: range divided by time of flight

(M1)

A1: Correct speed. Accept 10.9 or 10.85.

(A1)

2

[16]

Q25.

(a) $s = \frac{1}{2} \times 10 \times 4 + 10 \times 4 + \frac{1}{2} \times (4+7) \times 10 + \frac{1}{2} \times 7 \times 10$

(= 20 + 40 + 55 + 35)

M1: Any one term correct.
M1: A second term correct.
A1: Correct expression for total distance.

M1M1A1

= 150 m

A1: Total distance correct.

A1

OR

$s = \frac{1}{2} \times (10 + 20) \times 4 + \frac{1}{2} \times (4 + 7) \times 10 + \frac{1}{2} \times 7 \times 10$

(= 60 + 55 + 35)

= 150 m

(M1M1A1)
(A1)

OR

$s = \frac{1}{2} \times 10 \times 4 + 10 \times 4 + 10 \times 4 + \frac{1}{2} \times 10 \times 3 + \frac{1}{2} \times 7 \times 10$

(= 20 + 40 + 40 + 15 + 35)

= 150 m

(M1M1A1)
(A1)

4

(b) Average Speed = $\frac{150}{40}$ = 3.75 m s^{-1}

M1: Their total distance divided by 40.

M1

A1F: Correct average speed based on their distance from part (a). Must be correct to three or more significant figures.

A1F

2

(c) $a = \frac{4}{10}$ = 0.4 m s^{-2}

M1: Any division involving the numbers 10 and 4.

M1

A1: Correct acceleration. CAO

A1

***Note on use of constant acceleration equations**: award M1 for correct equation with correct values and A1 for correct final answer.*

2

(d) F = 200000 × 0.4 = 80000 N

M1: Multiplication of, 2 × 10^{n}, for any integer n, by candidate's acceleration from part (c).

A1F: Correct force based on their answer to part (c) multiplied by 200000.

M1A1F

Note: use of a = 2.5 gives 500000 N

Accept 80 kN

2

[10]

Q26.

(a) (i) P – 500 = 2200 × 0.8

P = 1760 + 500

M1: Equation of motion for car and caravan as a single body. Must see 2200 (or 1200 + 1000) multiplied by 0.8, and 500 (or 200 + 300). Allow sign errors.

A1: Correct equation.

M1A1

= 2260

A1: Correct value for P.

A1

(Award full marks for: (P =) 1760 + 500 = 2260 or similar to obtain correct final answer.)

OR (If finding the tension first)

$P - 1100 - 200 = 1200 \times 0.8$
$P = 960 + 1100 + 200$

M1: Equation of motion for car with their value for the tension.
Must see 1200 multiplied by 0.8, 200 and their tension.
Allow sign errors.
A1: Correct equation.

(M1A1)

= 2260

A1: Correct value for P.

(A1)

(Award full marks for: (P =) 960 + 200 + 1100 = 2260 or similar to obtain correct final answer.)

3

(ii) $T - 300 = 1000 \times 0.8$
$T = 300 + 800$

M1: Equation of motion for caravan.
Must see 300 and 1000 multiplied by 0.8.
Allow sign errors.
A1: Correct equation.

M1A1

= 1100

A1: Correct tension. CAO

A1

OR
$2260 - 200 - T = 1200 \times 0.8$
$T = 2260 - 200 - 960$

M1: Equation of motion for car. Must see 2260 (or candidate's P), 200 and 1200 multiplied by 0.8. Allow sign errors.
A1: Correct equation.

(M1A1)

= 1100 N

A1: Correct tension. CAO

(A1)

If candidates find tension first it must be stated in part (a)(ii) to gain any marks.

The working does not have to be repeated if seen in part (a)(i).

3

(b) (i) $15 = 7 + 0.8t$

M1: Use of a constant acceleration equation to find t, with 7, 15 and 0.8.

A1: Correct equation.

M1A1

$t = \dfrac{15-7}{0.8}$ = 10 seconds

A1: Correct time. CAO

A1

3

(ii) $15^2 = 7^2 + 2 \times 0.8s$

M1: Use of a constant acceleration equation to find s, with 7, 15 and 0.8.

A1: Correct equation

M1A1

$s = \dfrac{15^2-7^2}{1.6}$ = 110 m

A1: Correct distance. CAO

A1

OR

$s = \frac{1}{2}(7 + 15) \times 10 = 110$m

M1: Use of a constant acceleration equation to find s, with 7, 15 and candidate's time.

A1F: Correct equation.

A1F: Correct distance.

(M1A1F)
(A1F)

OR

$s = 7 \times 10 + \frac{1}{2} \times 0.8 \times 10^2 = 110$ m

M1: Use of a constant acceleration equation to find s, with 7, 0.8 and candidate's time.

A1F: Correct equation.

A1F: Correct distance.

(M1A1F)
(A1F)

If candidates find distance first it must be stated in part (b)(ii) to gain any marks.

The working does not have to be repeated if seen in part (b)(i).

3

(c) Resistance forces <u>vary with speed</u> (or velocity)
OR Speed (or velocity) changes (or increases)
OR It accelerates

B1: Correct explanation. Must not mention friction in main argument

B1

1

[13]

Q27.

(a) **v** = (4**i** + 0.5**j**) + (–0.4**i** + 0.2**j**)*t*

*M1: Use of constant acceleration equation to find **v** with **u** ≠ 0**i** + 0**j***

*A1: Correct **v**.*

(Could be done as a column vector.)

M1A1

2

(b) (i) **v** = (4**i** + 0.5**j**) + (– 0.4**i** + 0.2**j**) × 22.5

M1: Substitution of 22.5 into their expression for the elocity, even if no marks awarded in part (a).

M1

= –5**i** + 5**j**

A1: Correct velocity CAO

(Could be done using column vectors.)

A1

2

(ii) North-west

B1: Correct statement of direction. Accept 315°.
Must follow from correct answer to (b)(i).

B1

1

(c) ($\mathbf{v}$ =)$(4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

B1: Grouping ***i*** *and* ***j*** *components at some point in the solution.*
(Could be done using column vectors.)
Allow $5 = (4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

B1

$5^2 = (4 - 0.4t)^2 + (0.5 + 0.2t)^2$

M1: Seeing both components of their velocity squared and added

A1: Correct equation. (Condone including ***i*** *and* ***j****.)*
For example: $5 = (4 - 0.4t)\mathbf{i}^2 + (0.5 + 0.2t)\mathbf{j}^2$ *scores B1M1A0*
$5^2 = (4 - 0.4t)\mathbf{i}^2 + (0.5 + 0.2t)\mathbf{j}^2$ *scores B1M1A1*

M1A1

$0.2t^2 - 3t - 8.75 = 0$

A1: Any correct simplified quadratic equation, with exactly three terms.

A1

$t^2 - 15t - 43.75 = 0$

$t = 17.5$ or $t = -2.5$

dM1: Solving the quadratic equation.

(Allow one substitution error in correctly quoted formula)
Candidates with an incorrect quadratic equation must show method to get dM1.

dM1

$t = 17.5$

A1: Correct positive solution stated.

A1

6

[11]

Q28.

(a) $12\sin 30° t - 4.9t^2 = -0.5$

$4.9t^2 - 12\sin30°t - 0.5 = 0$

M1: Three term equation for vertical motion, with ±g, ±0.5 (or ±1 and ±1.5) and 12sin 30°t or 12cos30°t.

A1: Correct terms. (one must be equivalent to ±0.5)

A1: Correct signs.

M1A1A1

t = 1.30281...*or* –0.078323...

dM1: Solving the quadratic to find t.

Must see use of quadratic equation formula or can be implied by seeing 1.303 or 1.302 or similar.

dM1

t = 1.30 seconds (to 3sf)   AG

A1: Correct time from correct working. Must see more than 3 significant figures in candidate's working before the final answer or two correct solutions to the quadratic (eg 1.3 and –0.08).

Accept 1.3

A1

OR

time up = 0.6122

time down = 0.6122 + 0.0783 = 0.6905

total time = 0.6122 + 0.6905 = 1.30 (to 3sf)

M1: Adding time up to time down having used a quadratic.

A1: 0.6122

dM1: Finding time down with a quadratic

A1: 0.6905

A1: Correct answer

Accept 1.3

(M1A1dM1A1A1)

OR

–6.767 = 12 sin 30° – *gt*

M1:Forms an equation to find t having found v first

A1: Correct terms

A1: Correct signs

(M1A1A1)

$$t = \frac{12\sin 30° + 6.767}{g} = 1.30281 = 1.30 \text{ (to 3sf)}$$

dM1: Solving for t

A1: Correct time from correct working. Must see *more than 3 significant figures in candidate's working before the final answer.*

Accept 1.3

(dM1A1)

5

(b) 12cos30° 1.303 = 13.5 m

M1: Finding horizontal displacement using 1.30 (or better) and 12cos30°.

Do not allow 12sin30°.

A1: Correct distance. AWRT 13.5.

M1A1

2

(c) v_y = 12sin30° – 9.8 1.3028 (= –6.767)

M1: Finding vertical component of velocity or velocity squared at impact.

Must include 12sin30° or 12cos30 and ±g

A1: Correct expression for vertical component. May have 1.3 or 1.30 instead of 1.3028. (Accept +6.767 or similar)

M1A1

$v = \sqrt{(12cos30°)^2 + (-6.767)^2}$ = 12.4 m s^{-1}

dM1: Finding speed from two components. May use 6.74.

A1: Correct speed. Allow 12.3 or AWRT 12.4.

Note using g = 9.81 still gives 12.4.

dM1A1

4

(d) tanθ = $\frac{6.767}{12cos30°}$

M1: Trigonometric equation to find angle.

Can only be those shown opposite or described below. For tan, fraction can be inverted. For sin, 10.4 can be used instead of 6.767. For cos, 6.767 can be used instead of 10.4. Can use their values from part (c) (eg 6.74 or 6.77).

M1

θ = 33.1°

A1F: Correct angle. Accept AWRT 33°.

A1F

OR

$\sin\theta = \frac{6.767}{12.4}$

$\theta = 33.1°$

OR

$\cos\theta = \frac{10.4}{12.4}$

$\theta = 33.1°$

Follow though vertical component or final speed from part (c).

2

(e) The weight is the only force acting.

OR

No air resistance.

B1: Appropriate assumption.

B1

1

[14]

Q29.

(a) (i) $0.6^2 = 0^2 + 2a \times 0.9$

M1: Correct use of constant acceleration equation with u = 0 to find a.

A1: Correct equation.

M1A1

$a = \frac{0.6^2}{1.8} = 0.2 \text{ ms}^{-2}$ **AG**

A1: Correct a but some intermediate working must be seen.

A1

Note that $0^2 = 0.6^2 + 2a \times 0.9$

Scores M0A0A0

Verification methods require a conclusion for full marks to be awarded.

Condone seeing just the second line of working.

3

(ii) $0.9 = \frac{1}{2}(0 + 0.6)t$

M1: Correct use of constant acceleration equation

with u = 0 (and a = 0.2 if needed) to find t.

M1

$t = \frac{0.9}{0.3} = 3$ seconds

A1: Correct time.

A1

Note: Do not penalise 0.9 = $\frac{1}{2}$ (0.6 + 0)t in the first method.

OR

$0.6 = 0 + 0.2t$

(M1)

$t = \frac{0.6}{0.2} = 3$ seconds

(A1)

Note: 0 = 0.6 + 0.2t scores M0A0 in the second method.

OR

$0.9 = \frac{1}{2}\,0.2t^2$

(M1)

$t = 3$ seconds

(A1)

2

(b) $T - 800 \times 9.8 = 800 \times 0.2$

M1: Three term equation of motion. Must have these three terms but can have incorrect signs. Must use a = 0.2

A1: Correct equation with correct signs.

(Allow 800g)

M1A1

$T = 7840 + 160 = 8000$ N

A1: Correct tension.

A1

Accept 8008 or 8010 from use of g = 9.81.

3

[8]

Q30.

(a) $s = 32 \times 12.5 = 400$ m

B1: Correct distance.

B1

1

(b) $1600 = \frac{1}{2}(32 + 18)t$

M1: Seeing 2000 – candidate's answer to part (a) calculated

dM1: Use of constant acceleration equation(s) to find t, with u = 32 and v = 18

M1dM1

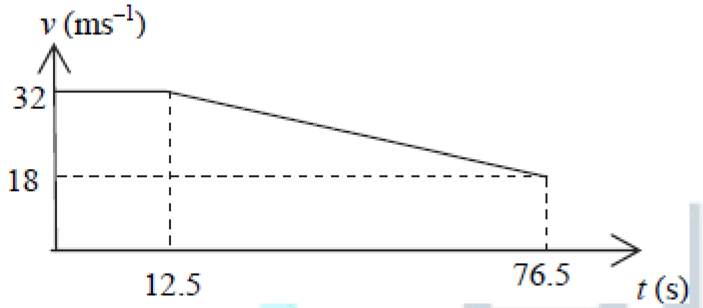
$t = \frac{1600}{25}$ = 64 seconds

A1: Correct time. Accept only 64

A1

3

(c)



B1: Shape of the graph.

B1

B1: Correct velocities (ie 18 and 32) on vertical axis.

B1

B1F: Correct times (ie 12.5 and 76.5) on the horizontal axis.

B1F

(Follow through incorrect answers to part (b)). Award marks for graph if seen in earlier parts.

3

(d) Average Speed = $\frac{2000}{12.5 + 64}$ = 26.1 ms^{-1}

M1: Use of 2000 over candidate's total time (not 64 or 12.5).

M1

A1F: Correct speed. AWRT 26.1.

FT candidate's answer to part (b) or (c).

A1F

2

[9]

Q31.

(a) $5g - T = 5a$

M1: Three term equation of motion with 5g or 49, 5a (not 5ga) and T.
A1: Correct equation.

M1A1

$T - 3g = 3a$

M1: Three term equation of motion with 3g or 29.4, 3a (not 3ga) and T.
A1: Correct equation.

M1A1

$2g = 8a$

$$a\left(=\frac{2g}{8}\right)=2.45\ \mathrm{ms}^{-2}$$

AG

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they then give their final answer as 2.45. If the final answer is –2.45 don't award the final A1 mark.

Special Case:

Whole String Method $2g = 8a$ and $a=\dfrac{2g}{8}=2.45$ *OE M1A1A1.*

5

(b) $T = 3 \times 9.8 + 3 \times 2.45$

M1: Substitution of a = 2.45 into a three term equation of motion to find the tension. Contains T, mg and ma where m = 3 or 5

M1

= 36.75

= 36.8 N (to 3 sf)

A1: Correct tension.
Accept 36.75 or 36.7

A1

2

(c) Light and Inextensible

B1: Light
B1: Inextensible (Allow inelastic or not stretchy)
Ignore irrelevant non-contradictory assumptions.

B1B1

2

(d) (i) $0.196 = \frac{1}{2} \times 2.45 \times t^2$

M1: Use of constant acceleration equation with $s = 0.196$, $u = 0$ and $a = 2.45$ to find t.

A1: Correct equation.

M1 A1

$$t = \sqrt{\frac{2 \times 0.196}{2.45}} = 0.4 \text{ seconds}$$

A1: Correct t

A1

3

(ii) $v^2 = 0^2 + 2 \times 2.45 \times 0.196$

$v = 0.98$

M1: Use of constant acceleration equation with $s = 0.196$, $a = 2.45$, $u = 0$ and candidate's time (as needed) to find v.

A1: Correct v.

M1A1

OR

$v = 0 + 2.45 \times 0.4 = 0.98 \text{ ms}^{-1}$

(M1A1)

OR

$$0.96 = \frac{1}{2}(0 + v) \times 0.4$$

(M1)

$v = 0.98 \text{ ms}^{-1}$

(A1)

2

[14]

Q32.

(a) $\mathbf{v} = (0.5\mathbf{i} + 0.375\mathbf{j}) \times 20$ $(=10\mathbf{i} + 7.5\mathbf{j})$

M1: Calculating velocity with $\boldsymbol{u} = 0\boldsymbol{i} + 0\boldsymbol{j}$ and $t = 20$.

A1: Correct expression for velocity.

M1A1

$$v = \sqrt{10^2 + 7.5^2} = 12.5 \text{ ms}^{-1}$$

dM1: Calculating speed.

A1: Correct speed.

dM1A1

4

(b) $\tan\theta = \frac{0.5}{0.375}$ or $\frac{10}{7.5}$ $\left(\text{or } \tan\theta = \frac{0.375}{0.5} \text{ or } \frac{7.5}{10}\right)$

M1: Using trig to find angle. Can be inverted as shown in brackets.

A1F: Correct trig expression with any correct equivalent fraction.

M1A1F

θ = 053°

A1: Correct angle to the nearest degree. Accept 53°.

A1

OR

$\cos\theta = \frac{7.5}{12.5}$ or $\frac{0.375}{0.625}$ $\left(\text{or } \cos\theta = \frac{10}{12.5}\right)$

(M1A1F)

θ = 053°

Note: For 37° award M1A0A0
But for 90 – 37 = 53° award M1A1A1.
For 127°, award M1A1A0

(A1)

OR

$\sin\theta = \frac{10}{12.5}$ or $\frac{0.5}{0.625}$ $\left(\text{or } \sin\theta = \frac{7.5}{12.5}\right)$

Note: 53.1° as final answer scores M1A1A0

(M1A1F)

θ = 053°

Condone finding angle from acceleration or position vector.

(A1)

3

(c) ($\mathbf{r}$ =) $\frac{1}{2}(0.5\mathbf{i} + 0.375\mathbf{j})\,t^2$ (= $0.25\,t^2\,\mathbf{i} + 0.1875t^2\,\mathbf{j}$)

M1: Finding an expression for position vector in terms of t.

A1: Correct position vector.

M1A1

$500^2 = (0.25\,t^2)^2 + (0.1875\,t^2)^2$

dM1: Using distance to form an equation for t.
A1: Correct equation.

dM1A1

$$t = 4\sqrt{\frac{500^2}{0.25^2 + 0.1875^2}} = 40 \text{ seconds}$$

A1: Correct time.

A1

OR

$a = 0.625$

M1: Finding magnitude of acceleration.
A1: Correct acceleration

(M1A1)

$$500 = \frac{1}{2}0.625t^2$$

dM1: Using distance to form an equation for t.
A1: Correct equation.

(dM1A1)

$t = 40$

A1: Correct time.

(A1)

OR

$400 = \frac{1}{2}\,0.5t^2$ *or* $300 = \frac{1}{2}\,0.375t^2$

M1: Working with one component.
A1: Correct distance (300 or 400)
A1: Correct equation.

(M1A1)
(A1)

$t^2 = 1600$

dM1: Solving for t.

(dM1)

$t = 40$

A1: Correct t.
Note: 500 ÷ 12.5 = 40 is not acceptable and scores 0

(A1)

5

[12]

Q33.

(a) $x = 40\cos\theta.t$

M1

$$y = -\frac{1}{2}(10)t^2 + 40\ \sin\theta.t$$

M1 A1

$$y = -\frac{1}{2}(10)(\frac{x}{40\ \cos\theta})^2 + 40\ \sin\theta.(\frac{x}{40\ \cos\theta})$$

Dependent on both M1s

m1

$$y = -\frac{x^2}{320\cos^2\theta} + x\tan\theta$$

$320\ y = -x^2(1 + \tan^2\theta) + 320x\tan\theta$

m1

$x^2\tan^2\theta - 320x\tan\theta + (x^2 + 320y) = 0$

Answer Given (Condone missing brackets)

A1

6

(b) (i) $150^2\tan^2\theta - 320(150)\tan\theta + (150^2 + 320 \times 8) = 0$

M1

$1125\tan^2\theta - 2400\tan\theta + 1253 = 0$

Correct quadratic

A1

$$\tan\theta = \frac{2400 \pm \sqrt{2400^2 - 4(1125)(1253)}}{2(1125)}$$

m1

$\tan\theta = 1.22$, 0.912

PI

A1F

$\theta = 50.7°$, $42.4°$

A1F

5

(ii) $\theta = 42.4°$

For the smaller angle

B1F

$t = \frac{150}{40\cos\theta}$ and cos 42.4 > cos 50.7

OE

E1

2

[13]

Q34.

(a) $0 = 4.9 - 9.8t$

M1: Use of constant acceleration equation to find t with v = 0.

A1: Correct equation

M1A1

$t = \frac{4.9}{9.8} = 0.5$ s

A1: Correct time

A1

3

(b) $0^2 = 4.9^2 - 2 \times 9.8s$

M1: Use of constant acceleration equation to find s with v = 0.

M1

$s = \frac{4.9^2}{19.6} = 1.225$ m

A1: Correct height.

A1

$H_{max} = 1.225 + 1 = 2.23$ m

A1: Correct total height.

A1

3

(c) 1 second

B1: Correct time.

B1

1

(d) 4.9 m s^{-1}

B1: Correct speed.

B1

1

[8]

Q35.

(a) $\mathbf{v} = 4.5\mathbf{i} + 2\mathbf{j}$

B1: Correct velocity

B1

1

(b) $\mathbf{v} = 4.5\mathbf{j}$

B1: Correct velocity

B1

1

(c) Due North

B1: Correct direction

B1

1

(d) $4.5\mathbf{j} = 4.5\mathbf{i} + 2\mathbf{j} + 5\mathbf{a}$

*M1: Use of a constant acceleration equation to find **a***

A1: Correct equation

M1A1

$\mathbf{a} = \dfrac{-4.5\mathbf{i} + 2.5\mathbf{j}}{5} = -0.9\mathbf{i} + 0.5\mathbf{j}$

A1: Correct acceleration

A1

3

(e) $\mathbf{F} = 250(-0.9\mathbf{i} + 0.5\mathbf{j}) = -225\mathbf{i} + 125\mathbf{j}$

M1: Using Newton's second law.

M1

$F = \sqrt{225^2 + 125^2} = 257 \text{ N}$

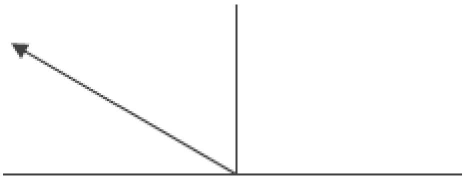
M1: Finding magnitude.

A1: Correct magnitude of the force

M1A1

3

(f)



B1: Diagram with arrow in the correct quadrant.

B1

$\tan^{-1}\left(\frac{125}{225}\right) = 29.1°$

M1: Use of trig to find an angle.

M1

A1: Correct angle from working.

A1

OR

$\tan^{-1}\left(\frac{0.5}{0.9}\right) = 29.1°$

$180 - 29.1 = 151°$

A1: Finding required angle correctly.

A1

4

[13]